Cryptography Exercises

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1 Statistical distance and information-theoretic encryption

Let A and B be two random variables taking values in a finite nonempty set $\mathcal{X}$. Define the *statistical distance*[1] *between A and B* as

$$\Delta(A\,;B) := \tfrac{1}{2} \cdot \sum_{a \in \mathcal{X}} |\Pr[A=a] - \Pr[B=a]|,$$

and *statistical distance of A from uniformity* as

$$d(A) := \Delta(A\,;U_{\mathcal{X}}),$$

where $U_{\mathcal{X}}$ is uniform over $\mathcal{X}$. The reference uniform distribution is always on the declared set; for a tuple, it is on the Cartesian product of the declared sets for its coordinates. Analogously, Δ can be defined as a distance between two *probability distributions* $\alpha, \beta : \mathcal{X} \to [0,1]$ as

$$\Delta(\alpha\,;\beta) := \Delta(A\,;B),$$

where A and B are random variables (taking values from $\mathcal{X}$) distributed according to α and β (respectively). A similar convention applies to the distance d from uniformity.

Randomized machines use private randomness independent of all external random choices.

Exercise 1.1 An alternative definition of statistical distance

Show that for every A and B we have

$$\Delta(A\,;B) = \sum_{a \in \mathcal{X}\colon \Pr[A=a] > \Pr[B=a]} \left(\Pr[A=a] - \Pr[B=a]\right).$$

Exercise 1.2 Statistical distance as a metric

Let Π be the set of all probability distributions over a finite nonempty set $\mathcal{X}$. Prove that Δ is a *metric* on this set, i.e., it satisfies the following axioms:

- *non-negativity:* for every $\alpha, \beta \in \Pi$ we have $\Delta(\alpha\,;\beta) \geq 0$,
- *identity of indiscernibles:* for every $\alpha, \beta \in \Pi$ we have that $\Delta(\alpha\,;\beta) = 0$ if and only if $\alpha = \beta$,
- *symmetry:* for every $\alpha, \beta \in \Pi$ we have that $\Delta(\alpha\,;\beta) = \Delta(\beta\,;\alpha)$, and
- *triangle inequality:* for every $\alpha, \beta, \gamma \in \Pi$ we have that $\Delta(\alpha\,;\gamma) \leq \Delta(\alpha\,;\beta) + \Delta(\beta\,;\gamma)$.

Exercise 1.3 Statistical distance as distinguishing advantage

Let A and B be two random variables taking values from some finite set $\mathcal{X}$. For a (possibly randomized) machine $\mathcal{D}$ that takes inputs from $\mathcal{X}$ and outputs a bit, define its *distinguishing advantage* between A and B as

$$\mathsf{Adv}^{\mathrm{dist}}_{\mathcal{D}}(A,B) := |\Pr[\mathcal{D}(A)=1] - \Pr[\mathcal{D}(B)=1]|.$$

[1] This is also known as the *total variation distance*, especially in the quantum information literature

Prove that for every computationally unbounded machine $\mathcal{D}$ we have

$$\mathsf{Adv}^{\mathrm{dist}}_{\mathcal{D}}(A, B) \leq \Delta(A\,;B). \tag{1}$$

For every A and B, construct $\mathcal{D}$ that achieves equality in Eq. (1). Show that the maximum is unchanged if the absolute value is omitted, and hence

$$\Delta(A\,;B) = \max_{\mathcal{D}} \left(\Pr[\mathcal{D}(A) = 1] - \Pr[\mathcal{D}(B) = 1]\right),$$

where the maximum is over all computationally unbounded machines that output a bit.

Exercise 1.4 Statistical distance and guessing advantage

Let α_0 and α_1 be two distributions over a finite nonempty set $\mathcal{X}$. Consider the following game played by a computationally unbounded, possibly randomized machine $\mathcal{M}$ (that knows α_0 and α_1):

1. A uniformly random bit $X \leftarrow \{0,1\}$ is chosen.
2. A value A is sampled according to distribution α_X and sent to $\mathcal{M}$.
3. $\mathcal{M}$ receives A and produces output $X' \in \{0,1\}$.

We say that $\mathcal{M}$ *won the game* if $X = X'$. Define its *guessing advantage* as

$$\mathsf{Adv}^{\mathrm{guess}}_{\mathcal{M}} := \left|\Pr[\mathcal{M} \text{ wins the game}] - \tfrac{1}{2}\right|.$$

Show that

$$\forall_{\mathcal{M}} \Pr[\mathcal{M} \text{ wins the game}] \leq \frac{1 + \Delta(\alpha_0\,;\alpha_1)}{2}. \tag{2}$$

For every α_0 and α_1, construct $\mathcal{M}$ that achieves equality in Eq. (2). By also considering the machine that complements its output, deduce that

$$\max_{\mathcal{M}} \mathsf{Adv}^{\mathrm{guess}}_{\mathcal{M}} = \tfrac{1}{2}\Delta(\alpha_0\,;\alpha_1).$$

Exercise 1.5 How function application changes statistical distance

Let A and B take values in a finite nonempty set $\mathcal{X}$, and let $\mathcal{Y}$ be a finite nonempty set. Show that for every function $f : \mathcal{X} \to \mathcal{Y}$ we have

$$\Delta(f(A)\,;f(B)) \leq \Delta(A\,;B).$$

Show that if f is an injection, then

$$\Delta(f(A)\,;f(B)) = \Delta(A\,;B).$$

Deduce that if f is a bijection from $\mathcal{X}$ onto $\mathcal{Y}$, then $d(f(A)) = d(A)$.

Exercise 1.6 One-time pad with imperfect randomness

Let $(\mathsf{Enc}, \mathsf{Dec})$ be the one-time pad encryption scheme for messages from $\{0,1\}^t$. Suppose a key K is chosen from $\{0,1\}^t$ according to some distribution α. Consider the following indistinguishability game played between a computationally unbounded adversary $\mathcal{A}$ and a challenger Ω:

1. $\mathcal{A}$ produces two messages $m_0, m_1 \in \{0,1\}^t$ and sends them to Ω.
2. Ω selects $X \leftarrow \{0,1\}$ uniformly at random, independently samples a fresh key K according to α, computes $c = \mathsf{Enc}(K, m_X)$, and sends c to $\mathcal{A}$.
3. $\mathcal{A}$ receives c and produces as output $X' \in \{0,1\}$.

We assume that $\mathcal{A}$ knows α. We say that $\mathcal{A}$ *won the game* if $X = X'$. Show that

$$\forall_{\mathcal{A}} \Pr[\mathcal{A} \text{ wins the game}] \leq \tfrac{1}{2} + d(\alpha).$$

Exercise 1.7 Conditional statistical distance

Let A and B take values in finite nonempty sets $\mathcal{X}$ and $\mathcal{Y}$, respectively. For each b with $\Pr[B = b] > 0$, let α_b be the conditional distribution of A given $B = b$, viewed as a distribution over $\mathcal{X}$. Define the *statistical distance of A from uniformity conditioned on B* as

$$d(A \mid B) := \sum_{b \in \mathcal{Y}:\, \Pr[B=b]>0} \Pr[B = b] \cdot d(\alpha_b).$$

Show that $d(A \mid B) = \Delta((A, B)\,; (U_{\mathcal{X}}, B))$, where $U_{\mathcal{X}}$ is uniform over $\mathcal{X}$ and independent of B.
Can you find a game-based interpretation of $d(A \mid B)$ similar to the one in Exercise Exercise 1.4?

Exercise 1.8 Conditioning does not decrease the distance from uniformity

Show that for every two random variables A and B taking values in finite nonempty sets, we have $d(A \mid B) \geq d(A)$.

2 Computational Indistinguishability

Distribution ensembles

A *distribution ensemble* is a family of probability distributions

$$\mathcal{A} = \{A_n\}_{n \in \mathbb{N}},$$

indexed by a security parameter n. We assume that A_n takes values in $\{0,1\}^{\ell(n)}$, where ℓ is polynomially bounded. As in the preceding section, we use the same notation for a distribution and a random variable distributed according to it.

Let $\mathcal{A} = \{A_n\}_{n \in \mathbb{N}}$ and $\mathcal{B} = \{B_n\}_{n \in \mathbb{N}}$ be two distribution ensembles over the same sets $\{0,1\}^{\ell(n)}$. For a probabilistic poly-time machine $\mathcal{D}$ that takes input $(1^n, x)$ and outputs a bit, define its *distinguishing advantage* as

$$\mathsf{Adv}^{\mathrm{dist}}_{\mathcal{D}}(n) := |\Pr[\mathcal{D}(1^n, A_n) = 1] - \Pr[\mathcal{D}(1^n, B_n) = 1]|.$$

The probabilities are taken over the random variables A_n and B_n, respectively, and the independent private randomness of $\mathcal{D}$.

A function $\mu : \mathbb{N} \to \mathbb{R}_{\geq 0}$ is *negligible* if for every constant $c > 0$ there exists $N_c \in \mathbb{N}$ such that for every $n \geq N_c$ we have $\mu(n) < n^{-c}$.

We say that $\mathcal{A}$ and $\mathcal{B}$ are *computationally indistinguishable*, denoted

$$\mathcal{A} \approx_c \mathcal{B},$$

if for every probabilistic poly-time machine $\mathcal{D}$ that outputs a bit, the function $\mathsf{Adv}^{\mathrm{dist}}_{\mathcal{D}}$ is negligible.

An equivalent formulation using samplers

A *sampler* is a probabilistic poly-time machine $\mathcal{S}$ that, on input 1^n, outputs a string of length $\ell(n)$, where ℓ is polynomially bounded.

Let $\mathcal{S}_0$ and $\mathcal{S}_1$ be two samplers with the same output length $\ell(n)$. They induce the distribution ensembles

$$\mathcal{A} = \{\mathcal{S}_0(1^n)\}_{n \in \mathbb{N}} \quad \text{and} \quad \mathcal{B} = \{\mathcal{S}_1(1^n)\}_{n \in \mathbb{N}}.$$

We say that $\mathcal{S}_0$ and $\mathcal{S}_1$ are *computationally indistinguishable* if their induced distribution ensembles are computationally indistinguishable. Equivalently, for every probabilistic poly-time machine $\mathcal{D}$ that outputs a bit, the function

$$\varepsilon_{\mathcal{D}}(n) := |\Pr[\mathcal{D}(1^n, \mathcal{S}_0(1^n)) = 1] - \Pr[\mathcal{D}(1^n, \mathcal{S}_1(1^n)) = 1]|$$

is negligible. The probabilities are taken over the independent private randomness of the sampler and $\mathcal{D}$. This formulation applies to ensembles generated by probabilistic poly-time samplers; general distribution ensembles need not admit such samplers.

Exercise 2.1 Guessing advantage and distinguishing advantage

Let $\mathcal{A} = \{A_n\}_{n\in\mathbb{N}}$ and $\mathcal{B} = \{B_n\}_{n\in\mathbb{N}}$ be two distribution ensembles. Consider the following game played between a probabilistic poly-time adversary $\mathcal{M}$ and a challenger Ω, with security parameter n:

1. Ω selects $X \leftarrow\!\!\$\ \{0,1\}$ uniformly at random.
2. If $X = 0$, Ω samples x according to A_n. Otherwise, it samples x according to B_n.
3. Ω sends $(1^n, x)$ to $\mathcal{M}$.
4. $\mathcal{M}$ receives $(1^n, x)$ and produces output $X' \in \{0,1\}$.

We say that $\mathcal{M}$ *won the game* if $X = X'$. Define its *guessing advantage* as

$$\mathsf{Adv}^{\mathrm{guess}}_{\mathcal{M}}(n) := \left|\Pr[\mathcal{M} \text{ wins the game}] - \tfrac{1}{2}\right|.$$

Show that

$$\mathsf{Adv}^{\mathrm{guess}}_{\mathcal{M}}(n) = \tfrac{1}{2}\mathsf{Adv}^{\mathrm{dist}}_{\mathcal{M}}(n),$$

where $\mathcal{M}$ is also viewed as a machine that outputs a bit in the distinguishing experiment. Deduce that $\mathcal{A} \approx_c \mathcal{B}$ if and only if every probabilistic poly-time adversary has negligible guessing advantage.

Exercise 2.2 Computational indistinguishability under efficient transformations

Let $\mathcal{A} = \{A_n\}_{n\in\mathbb{N}}$ and $\mathcal{B} = \{B_n\}_{n\in\mathbb{N}}$ be two distribution ensembles over $\{0,1\}^{\ell(n)}$. Let F be a probabilistic poly-time machine that, on input $(1^n, x)$ with $x \in \{0,1\}^{\ell(n)}$, outputs a string of length $m(n)$. Write $F_n(x)$ for the output of $F(1^n, x)$, including its private randomness.

(a) Suppose that $\mathcal{A} \approx_c \mathcal{B}$. Prove that

$$\{F_n(A_n)\}_{n\in\mathbb{N}} \approx_c \{F_n(B_n)\}_{n\in\mathbb{N}}.$$

Solution: Suppose, towards a contradiction, that there exists a probabilistic poly-time machine $\mathcal{D}$ whose distinguishing advantage between the transformed ensembles,

$$\varepsilon(n) := |\Pr[\mathcal{D}(1^n, F_n(A_n)) = 1] - \Pr[\mathcal{D}(1^n, F_n(B_n)) = 1]|,$$

is non-negligible. Construct a machine $\mathcal{D}'$ as follows.

Distinguisher $\mathcal{D}'$

On input $(1^n, x)$ do the following:

1. Compute $y := F(1^n, x)$ using fresh private randomness.
2. Output $\mathcal{D}(1^n, y)$.

Since F and $\mathcal{D}$ are probabilistic poly-time machines, so is $\mathcal{D}'$. Moreover,

$$\mathsf{Adv}^{\mathrm{dist}}_{\mathcal{D}'}(n) = |\Pr[\mathcal{D}(1^n, F_n(A_n)) = 1] - \Pr[\mathcal{D}(1^n, F_n(B_n)) = 1]| = \varepsilon(n).$$

Thus $\mathcal{D}'$ distinguishes $\mathcal{A}$ from $\mathcal{B}$ with non-negligible advantage, contradicting $\mathcal{A} \approx_c \mathcal{B}$.

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2. Show that the converse of part (a) is false, i.e., that

$$\{F_n(A_n)\}_{n\in\mathbb{N}} \approx_c \{F_n(B_n)\}_{n\in\mathbb{N}}$$

does not necessarily imply $\mathcal{A} \approx_c \mathcal{B}$.

Solution: Let $A_n = 0^n$ and $B_n = 1^n$ with probability 1. The poly-time machine $\mathcal{D}$ defined by

$$\mathcal{D}(1^n, x) := \begin{cases} 1 & \text{if } x = 0^n, \\ 0 & \text{otherwise} \end{cases}$$

has distinguishing advantage 1 for every $n \geq 1$. Hence $\mathcal{A} \not\approx_c \mathcal{B}$.

Now let $F(1^n, x) := 0^n$. This machine runs in polynomial time and satisfies $F_n(A_n) = 0^n = F_n(B_n)$. Thus the transformed distributions are identical for every n, and their ensembles are computationally indistinguishable.

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3. Suppose now that F is deterministic and, for every n, the function

$$f_n : \{0,1\}^{\ell(n)} \to \{0,1\}^{m(n)}$$

computed by $F(1^n, \cdot)$ is injective. Does

$$\{f_n(A_n)\}_{n\in\mathbb{N}} \approx_c \{f_n(B_n)\}_{n\in\mathbb{N}}$$

necessarily imply $\mathcal{A} \approx_c \mathcal{B}$? Prove your answer or provide a counterexample.

3 Symmetric encryption

Exercise 3.1 Negligible functions

Let $\varepsilon : \mathbb{N} \to \mathbb{R}$ be an arbitrary function. Split ε into two functions: ε_0 and ε_1 that are equal to ε on even and odd inputs, respectively. More precisely: for $b \in \{0,1\}$ let ε_b be the function defined for every $m \in \mathbb{N}$ as $\varepsilon_b(m) := \varepsilon(2m + b)$. Show that if both ε_0 and ε_1 are negligible, then so is ε.

Solution: To prove that ε is negligible, take an arbitrary $c \in \mathbb{Z}$. We need to show that there exists $N \in \mathbb{N}$ such that for every $n > N$, we have that $|\varepsilon(n)| \leq n^{-c}$. Since for ε_0 and ε_1 are negligible, thus there exist M_0 and M_1 such that for every $m > M_b$ we have that $\varepsilon_b(m) \leq m^{-c-1}$. Hence, for every $m > \max(M_0, M_1)$ we have that

$$|\varepsilon_0(m)| \leq m^{-c-1} \text{ and } |\varepsilon_1(m)| \leq m^{-c-1},$$

which can be rewritten as

$$|\varepsilon(2m)| \leq m^{-c-1} \text{ and } |\varepsilon(2m+1)| \leq m^{-c-1}.$$

This implies that for every $n > \max(2M_0, 2M_1)$ we have $|\varepsilon(n)| \leq \lfloor n/2 \rfloor^{-c-1}$. Set $N := \max(2M_0, 2M_1, 3^{c+1})$. We get

$$\begin{aligned} |\varepsilon(n)| &\leq \lfloor n/2 \rfloor^{-c-1} \\ &\leq (n/3)^{-c-1} \qquad (3) \\ &\leq n^{-c} \cdot 3^{c+1}/n, \qquad (4) \end{aligned}$$

where in Eq. (3) we used the fact that $\lfloor n/2 \rfloor \geq n/3$ (which holds for every $n \geq 2$) and in Eq. (4) we used the fact that $n > N \geq 3^{c+1}$.

Exercise 3.2 Noticeable functions

A function μ is noticeable iff there exists $c \in \mathbb{N}$ and $n_0 \in \mathbb{N}$ such that for every $n \geq n_0$ we have that $|\mu(n)| \geq n^{-c}$. Answer the following questions:

(a) Is every noticeable function non-negligible?
(b) Is every non-negligible function noticeable?

Exercise 3.3 One-way functions

Let f and g be one-way functions that are *length-preserving*, i.e., for every $x \in \{0,1\}^*$ we have that $|f(x)| = |g(x)| = |x|$. For h_i's defined as below, decide if they are one-way functions (if the answer is "yes" then give a proof, and otherwise provide a counterexample).

(a) $h_1(x) := f(x) \oplus g(x)$,
(b) $h_2(x) := f(g(x))$,
(c) $h_3(x) = f(x)||g(x)$,

Solution: The answer is "no", i.e., it *is* possible that f and g are one-way, while h_3 is not. Below, we construct an example of such f and g. Let $\alpha : \{0,1\}^* \to \{0,1\}^*$ be a length-preserving one-way function. We use the following notation: for a string $x = (x_1, \ldots, x_n)$ and $a, b \in \mathbb{N}$ (such that $1 \leq a \leq b \leq n$) let $x[a, \ldots, b]$ denote the string $(x_a, \ldots, x_b)$. For x length n define

$$f(x) := \alpha\left(x[1, \ldots, \lfloor n/2 \rfloor]\right) \,||\, x[\lfloor n/2 \rfloor + 1, \ldots, n]$$
$$g(x) := \alpha\left(x[\lfloor n/2 \rfloor + 1, \ldots, n]\right) \,||\, x[1, \ldots, \lfloor n/2 \rfloor].$$

For such f and g we have

$$h_3(x) = \alpha\left(x[1, \ldots, \lfloor n/2 \rfloor]\right) \,||\, x[\lfloor n/2 \rfloor + 1, \ldots, n] \,||\, \alpha\left(x[\lfloor n/2 \rfloor + 1, \ldots, n]\right) \,||\, x[1, \ldots, \lfloor n/2 \rfloor].$$

Since x is a part of the output of $h(x)$, thus function h_3 is *not* one-way. It remains to show that f and g are one-way. Since these functions are symmetric, it suffices to prove this fact just for one of them, say f. We show that f is one-way by contradiction. Suppose f is not one-way, i.e., there exists a poly-time adversary $\mathcal{A}^f$ such that ε defined as

$$\varepsilon(n) := \Pr\left[f(x') = y : x \leftarrow\!\$\ \{0,1\}^n;\ y := f(x);\ x' := \mathcal{A}^f(y)\right] \tag{5}$$

is non-negligible. For $b \in \{0,1\}$ let ε_b be a function defined for every $m \in \mathbb{N}$ as $\varepsilon_b(m) := \varepsilon(2m + b)$. By Exercise 3.1 it cannot be the case that both ε_0 and ε_1 are negligible. Without loss of generality assume that ε_0 is non-negligible. We will construct an adversary $\mathcal{A}^\alpha$ that with non-negligible probability inverts α. More precisely, $\mathcal{A}^\alpha$ will be such that

$$\Pr[\alpha(w') = v : w \leftarrow\!\$\ \{0,1\}^m;\ v := \alpha(w);\ w' := \mathcal{A}^\alpha(v)] = \varepsilon_0(m). \tag{6}$$

The adversary $\mathcal{A}^\alpha$ is presented below.

Adversary $\mathcal{A}^\alpha$

On input v of length m do the following:

1. sample $z \leftarrow\!\$\ \{0,1\}^m$,
2. let $q := \mathcal{A}^f(v \,||\, z)$,
3. output $q[1, \ldots, m]$.

It is easy to see that if $v := \alpha(w)$, where $w \leftarrow\!\$\ \{0,1\}^m$, then the distribution of the random variable $v \,||\, z$ in Step 2 of the above procedure is identical to the distribution of y in the experiment from Eq. (5) with $n = 2m$. This means that the probability that $\mathcal{A}^f$ in Step 2 outputs q such that $f(q) = v||z$ is equal to $\varepsilon(2m)$, which, in turn, is equal to $\varepsilon_0(n)$. By the definition of f we have that $f(q) = v \,||\, z$ only if $\alpha(q[1, \ldots, m]) = v$. Hence Eq. (6) is proven. Note that the adversary $\mathcal{A}^\alpha$ just samples z and runs $\mathcal{A}^f$ once. Since we assumed that $\mathcal{A}^f$ is poly-time, thus the running time of $\mathcal{A}^\alpha$ is also polynomial. Hence the proof that f is one-way is finished. The proof for g is analogous to the one for f.

(d) $h_4(x) := x \oplus f(x)$.